

HBS formation as a second-order phase transition

A Landau functional for SCIANTIX

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SCIANTIX status

The model

Calibration

Results

Model applied to SCIANTIX

Outlook: from a local functional to a phase field

SCIANTIX status

SCIANTIX had three formation models for the restructured volume fraction

option	form	driven by
1	KJMA, Barani (2020) ¹	effective burnup
2	KJMA + $bu_{inc} = 15$, Biswas and Aagesen (2025) ²	effective burnup
3	KJMA on ρ_d , Veshchunov ³ + Zullo (2026)	local burnup

All three are **transformation kinetics**: a sigmoid fitted in burnup. None of them produces the **mean misorientation** Θ of the subgrains, or the **subgrain radius** r_n .

¹ Barani et al., *J. Nucl. Mater.* 539 (2020) 152296.

² Biswas and Aagesen, *Comput. Mater. Sci.* 258 (2025) 114052, Eq. 45.

³ Veshchunov and Shestak, *J. Nucl. Mater.* 384 (2009) 12–18.

The data this model is built on

paper	technique	Θ	X	r_n
Zacharie-Aubrun (2022), standard UO_2	EBSD	27	27	14
Zacharie-Aubrun (2022), Cr-doped	EBSD	11	11	8
Onofri (2025)	EBSD, no restructuring ($\Theta = 0$)	14	–	–
Gerczak (2018)	image analysis (Barani 2020), $d_A/2$	–	5	3
Noirot (2015)	Xe-depleted area	–	6	–
total		52	49	25

$$\Theta = [A_{\text{Mis}}(f_1 - f_{10}) + 10f_{10}]/100;$$

$$X = f_{10}, \text{ area fraction above } 10^\circ;$$

$$r_n = \text{ECD50 \%}/2.$$

1° is the lower bin edge of the EBSD data.

What KJMA is, and what it assumes

Nuclei appear *at random* in the untransformed material and grow *isotropically*; count the transformed volume as if the growing regions could overlap freely, the **extended volume** V_e , and the overlap correction is statistical:

$$dX = (1 - X) \frac{dV_e}{V} \implies X = 1 - \exp(-K t^n)$$

What you get

K and n , and a sigmoid. n carries the dimensionality of growth and the nucleation regime.

What it assumes

Randomly placed nuclei, a growth law independent of where you are in the transformation, and a *time dependence*.

The time dependence in the KJMA model

- 1 **Effective burnup**, $n = 3.54$, $K = 2.77 \times 10^{-7}$, fitted on restructured-fraction data.
- 2 Same, with an **incubation burnup** $bu_{inc} = 15 \text{ MWd/kgHM}$ below which nothing happens. Biswas & Aagesen get it from a balance between the energy stored in dislocations and the energy of forming a subgrain used to shift the origin of an empirical curve rather than to build the curve.
- 3 KJMA not on burnup but on the **dislocation density** $\rho_d(bu, T)$, with a threshold $\rho_{crit} = 6 \times 10^{14} \text{ m}^{-2}$. The first to make the driver a **microstructural state variable** and the only one that feels temperature.

What none of them can do

They are calibrated on X alone. There is no misorientation angle neither subgrain size.

Why a phase transition could be the right shape for this problem

The HBS forms *continuously*

Not by nucleating new grains, but by low-angle boundaries absorbing dislocations until their misorientation drifts through the 10° HAGB threshold. In the metallurgical literature this is *continuous* dynamic recrystallization, and it is a well-posed alternative to the nucleation-and-growth picture KJMA describes.⁵

A continuous transformation has an order parameter

If a single continuous quantity distinguishes the two states, Landau's argument applies: expand the free energy in it, and keep only the terms symmetry allows.⁶

$$F(\eta) = C_0 + C_2\eta^2 + C_4\eta^4$$

Odd powers are forbidden here since the energy of a boundary cannot depend on the *sign* of the misorientation across it.

C_2 changing sign produces the transition between a null order parameter and a non-zero one, i.e. the development of misorientation.

⁵Gourdet & Montheillet, *Acta Mater.* 51 (2003) 2685; Humphreys & Hatherly (2017), chs. 6 and 10.

⁶Landau (1937); Landau & Lifshitz, *Statistical Physics I*, §143.

Landau against KJMA, side by side

	KJMA (options 1–3)	Landau (option 4)
what is solved	a <i>kinetic</i> law along a path	the <i>equilibrium</i> of a state
driver	burnup or ρ threshold	$\rho(bu)$ through C_2
free parameters	K , n fitted on X	β , k , ρ_c fitted jointly on Θ , X , r_n
predicts X	by construction	yes , lever rule, no extra parameter
predicts Θ , r_n	no	yes
reversible?	no, by construction	yes, see below

The price of using an equilibrium theory

A drop in ρ_{tot} would un-restructure the fuel. Restructuring is irreversible, so the implementation carries a **monotonic lock** on Θ . However, this is easy if one considers the dislocation density generated by irradiation, and eventually includes the annealing phenomena within the evolution of the Landau free energy.

The model

HBS as a second-order phase transition

Order parameter

$$\eta = \frac{\theta}{\theta_{\max}}, \quad \theta_{\max} \equiv \theta_{\text{HAGB}} = 10^\circ$$

the mean misorientation normalised to the LAGB/HAGB boundary.

External condition: the local burnup, entering *only* through the dislocation density

$$\rho_{\text{tot}}(bu) = 10^{2.2 \times 10^{-2} bu + 13.8}$$

Nogita & Une (1994)

Free energy

$$F(bu, \eta) = C_0 + C_2 \eta^2 + C_4 \eta^4$$

Only even powers: the energy cannot depend on the *sign* of a misorientation.

Minimising F gives Θ ; the same wall geometry gives r_n ; the lever rule gives X .

Three adjustable parameters

β (wall geometry), k (sweeping), ρ_c (strain-field cut-off).

The geometric wall

A low-angle boundary *is* a wall of dislocations.

1. Spacing sets the angle $\theta = b/d$

so a wall of n dislocation families carries a line length per unit *wall area*

$$L_A = n\theta/b$$

2. Wall area per unit volume or low angle grain boundary surface

$$S/V = 3\sqrt{\rho_{\text{LAGB}}} / \beta$$

β : the one geometric parameter (Rest & Hofman 2000).
 $n = 2$, Gourdet & Montheillet, not fitted (between 1-3).

Close the two on each other:

$$\rho_{\text{LAGB}} = L_A \cdot S/V$$

$$\Rightarrow \sqrt{\rho_{\text{LAGB}}} = \frac{3n\theta}{\beta b}$$

Everything follows from θ alone

$$\rho_{\text{LAGB}} = \left(\frac{3n\theta}{\beta b} \right)^2, \quad r_n \simeq \frac{\beta^2 b}{6n\theta}$$

The dislocation balance

Every dislocation is in exactly one of three states, and they must add up to ρ_{tot} :

$$\underbrace{\rho_{\text{ord}} = \rho_{\text{LAGB}}^{\text{max}} \eta^2}_{\text{condensed into low-angle walls}} + \underbrace{\rho_{\text{swept}} = (\rho_{\text{tot}} - \rho_{\text{ord}}) \frac{\Delta R}{R}}_{\text{annihilated by moving boundaries}} + \underbrace{\rho_{\text{free}}}_{\text{still a random array}} = \rho_{\text{tot}}$$

Each population carries the line energy of the state it is in:

$$F = \rho_{\text{free}} A_1 G b^2 + \rho_{\text{ord}} A_2 G b^2, \quad A_i = \frac{f(\nu)}{4\pi} \ln \frac{R_i}{b}$$

free ones screened at ρ_c , wall ones by the wall itself at ρ_{tot} ; the swept ones carry nothing.

$$C_0 = \rho_{\text{tot}} A_1 G b^2, \quad C_4 = \rho_{\text{LAGB}}^{\text{max}} \left. \frac{\Delta R}{R} \right|_{\text{max}} A_1 G b^2$$

$$C_2 = \underbrace{\rho_{\text{LAGB}}^{\text{max}} (A_2 - A_1) G b^2}_{\text{walls: the screening gain}} - \underbrace{\rho_{\text{tot}} \left. \frac{\Delta R}{R} \right|_{\text{max}} A_1 G b^2}_{\text{sweep}}$$

The engine of the transition is $A_2 - A_1$. As burnup rises, ρ_{tot} rises, the screening length due to wall formation shrinks, A_2 falls below A_1 , and it becomes cheaper to store dislocations in walls than to leave them random.

Equilibrium

$$\frac{\partial F}{\partial \eta} = 0 \implies \eta^2 = -\frac{C_2}{2C_4}, \quad \text{clipped at 0}$$

$$\eta = 0 \text{ wherever } C_2 \geq 0.$$

Condition on the minimisation problem: dislocation balance

The walls cannot hold more dislocations than exist, so $\rho_{\text{ord}} \leq \rho_{\text{tot}}$. The equilibrium is the minimum of F on the **admissible interval**, not the free stationary point:

Eq. (7b) — admissibility

$$\eta_{\text{bal}} = \sqrt{\min\left(\frac{\rho_{\text{tot}}}{\rho_{\text{LAGB}}^{\text{max}}}, 1\right)}, \quad \eta_{\text{eq}} = \min\left(\sqrt{-\frac{C_2}{2C_4}}, \eta_{\text{bal}}\right)$$

With the current calibration the bound **never binds** (0 of the 41 standard EBSD points, no burnup at 600 K): it stays in the model as a physical constraint, not as an active branch.

Calibration

The objective and the three data sets

Joint fit of β, k, ρ_c ($n = 2$ fixed) on all three observables:

$$J = \frac{\langle \Delta \Theta^2 \rangle_w}{\text{var } \Theta} + w_r \frac{\langle \Delta r^2 \rangle_w}{\text{var } r} + w_X \frac{\langle \Delta X^2 \rangle_w}{\text{var } X}$$

dimensionless;

$\langle \cdot \rangle_w$: average with a weight w_i per point.

Two questions:

1. **which weights** w_r, w_X ?
2. **which data set**?

set	points, weights
A	Zacharie-Aubrun + Onofri, standard UO_2 ; $w_i = 1$
B	all four papers; $w_i = 1$
C	as B, $w_i = \text{rank factor} \times \text{relevance}/3$

rank factor: A, B $\rightarrow 1$; $\text{U}^* \rightarrow 0.5$; $\text{U} \rightarrow 0.25$.

Where the data come from: quality rank $\times$ relevance

$$w_i = \underbrace{f(\text{Rose rank})}_{\text{how well it was measured}} \times \underbrace{\text{relevance}/3}_{\text{how close to our application}}$$

1. Rose rank — M.A. Rose, ANL/CFCT-22/26 (2022) (MSTDB-TP), re-worded for post-irradiation microscopy. Six aspects marked H/M/I: *method, calibration, sample characterisation, irradiation conditions, measurement precision, verifiability*.

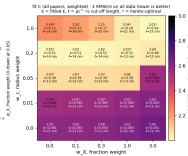
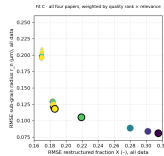
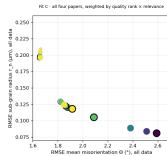
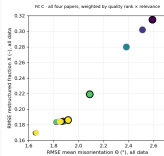
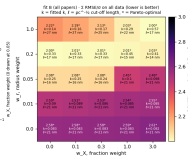
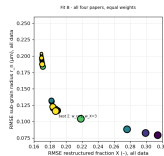
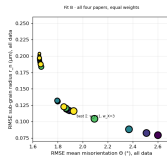
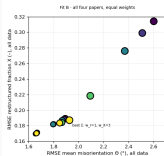
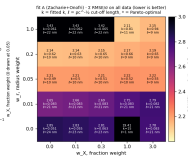
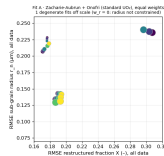
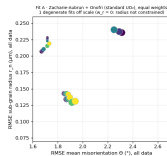
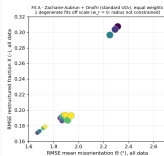
rank	assigned when	f
A	all H/M <i>and</i> verifiability H	1
B	≥ 2 H/M <i>and</i> verifiab. H/M	1
U*	all but verifiability H/M, unc. unusable	0.5
U	uncertainties bar quantitative use	0.25

2. Relevance (1–3).

sample class	relevance
PWR standard UO ₂	3
PWR Cr-doped UO ₂	2
Halden disc, std	2
Halden disc, large grain	1

Choosing the weights: the calibration front

Calibration front of the Landau model: one fit per data set, w, r, μ errors on ALL the data
marker colour = w, X , marker size = w, r (0, 0.01, 0.05, 0.2, 1); black ring = not dominated by any other fit (all data sets pooled)



The calibrated parameters

Set C, $w_r = 0.2$, $w_X = 1$

all four papers, each point weighted by
quality rank $\times$ relevance

n (not fitted)	2
β	26.682
k	0.5102
ρ_c	$5.741\text{e}15 \text{ m}^{-2}$

In sample, all four papers

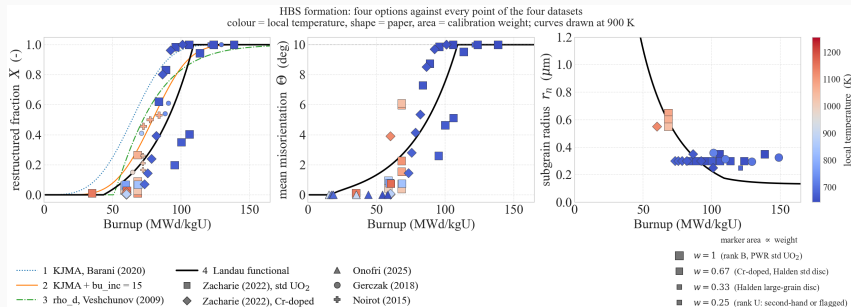
weighted, with the unweighted beside it

	N	RMSE / R^2	plain
Θ	52	$1.896^\circ / 0.733$	$1.864 / 0.749$
X	49	$0.194 / 0.723$	$0.184 / 0.744$
r_n	26	$0.122 \mu\text{m} / \mathbf{0.02}$	$0.122 / \mathbf{-0.04}$

- X is in the objective.
- The size is still the price: $R^2 \approx 0$ on all 26 radii — standard UO_2 alone is $+0.47$, dragged down by the Cr-doped (-1.41) .

Results

Comparison with the other options and with all the data



Restructured fraction X , all $N = 49$ (weighted; plain in grey)

option	RMSE	R^2
1 KJMA	0.389	-0.11
2 KJMA + bu_{inc}	0.217	+0.65
3 ρ_d	0.232	+0.60
4 Landau	0.194	+0.72

plain: 0.380 / 0.204 / 0.227 / 0.184

Option 4 only: Θ 1.90° / +0.73
($N = 52$), r_n $0.122 \mu\text{m}$ / +0.02
($N = 26$). Fit C, the same weights.

Against the three models SCIANTIX already had

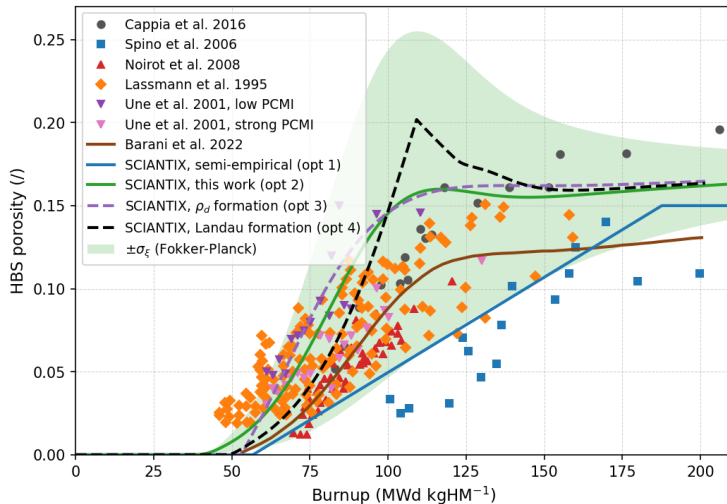
Burnup [MWd/kgU] at which α_r first reaches:

threshold	1 KJMA	2 KJMA+ bu_{inc}	3 ρ_d	4 Landau
1 %	19.5	34.5	51.2	45.0
10 %	37.8	52.8	54.8	58.6
50 %	64.2	79.3	72.3	89.2
90 %	90.1	105.2	112.3	105.1
99 %	109.7	124.6	161.7	107.8

- Option 4 starts **between options 2 and 3** (45 MWd/kgU) but rises more slowly: its 50 % crossing is the **latest of the four**.
- It still **saturates**: Θ hits the 10° cap and $X = 1$ at 108 MWd/kgU, whereas KJMA approaches 1 asymptotically.
- The 1–99 % span is **63 MWd/kgU**, against 90 for option 1 and 110 for option 3.

Model applied to SCIANTIX

Does the new formation model disturb the porosity model?



Largest deviation from option 2 (test_U02HBS):

**+0.044 absolute at
95.2 MWd/kgHM
(+28 %).** Option 4

overshoots to 0.180 at
108 MWd/kgU — its
saturation — and dips by
14.5 % after it. All four
options still end within
0.37 % of the same
porosity.

Outlook: from a local
functional to a phase field

What a local functional cannot give -> phase-field models

Problems of the Landau-like local functional

- no spatial quantities;
- no kinetics: equilibrium is instantaneous.

family	property	application/ref
KWC	two fields (ϕ, θ), allows subgrain rotation; singular diffusion equations	Nucleation via subgrain coalescence, Muramatsu (2014)
MPF	One order parameter per grain; stable time increments; rotation ignored	Subgrain growth, Khajezade <i>et al.</i> (2024)
HMP	Removes nonphysical linear gradient term; handles inclination-dependent GB energy	Coupled Cosserat crystal plasticity, Tandogan <i>et al.</i> (2024); dislocation-driven nucleation, Tandogan <i>et al.</i> (2026)
PFC	Atomic-resolution density field; diffusive timescales	Interface morphology development during solidification, Nit-tala (2025)

Nucleation is usually **seeded ad hoc** at a critical density or stress; spontaneous nucleation from lattice-curvature gradients is obtained when coupling KWC with Cosserat crystal plasticity (Ghiglione, 2024), and from stored dislocations at grain boundaries when coupling HMP with it (Tandogan, 2026).

Muramatsu *et al.* (2014): a KWC phase field for recrystallization

The functional⁷ — two fields, ϕ and θ :

$$\mathcal{F} = \int_V \left[f(\phi) + \frac{\alpha^2}{2} |\nabla \phi|^2 + s g(\phi) |\nabla \theta| + \frac{\epsilon^2}{2} h(\phi) |\nabla \theta|^2 \right] dV$$

with $g(\phi) = h(\phi) = \phi^2$, relaxed by $\dot{\phi} = -M_\phi \delta \mathcal{F} / \delta \phi$ and $\dot{\theta} = -M_\theta \delta \mathcal{F} / \delta \theta$.

$\phi = 1$ inside a subgrain; θ is the **lattice orientation**.

The *singular* $|\nabla \theta|$ term gives a boundary an energy growing with misorientation.

The local part (Eq. 7, $f_m = 0$):

$$f(\phi; E_s) = a_{ub}(1 - p(\phi)) E_s + a_{wg} W_b q(\phi)$$

$$q(\phi) = \phi^2(1 - \phi)^2$$

$$p(\phi) = \phi^3(10 - 15\phi + 6\phi^2)$$

- q : the **double well**, original matrix against new grains; its height $a_{wg} W_b$ sets the boundary energy;
- p : is a sigmoid-like function where $p'(0) = p'(1) = 0$ therefore it *tilts* the well without moving its two minima;
- E_s : the **stored dislocation energy**, the only driving force, supplied from outside the model.

How the paper nucleates: a subgrain group enclosed by high-angle ($> 10^\circ$) boundaries is *selected* as a nucleus candidate and grows after an *incubation period* — the criterion is **imposed**, not produced by the free energy.

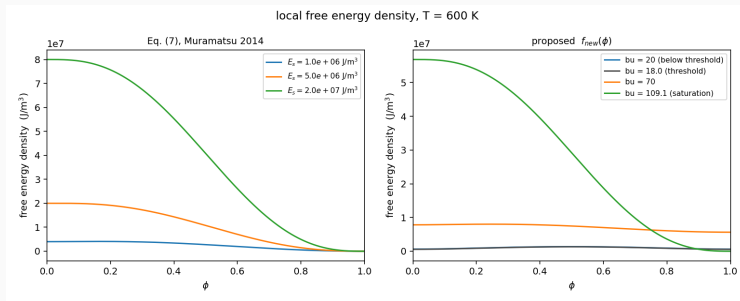
The prototype: the Landau model supplies the driving force

Keep $q(\phi)$ and $p(\phi)$ exactly as above, and replace the externally supplied E_s by what the Landau model:

$$f_{\text{Mur}}(\phi; E_s) = a_{ub}(1 - p(\phi))E_s + a_{wg}W_b q(\phi)$$

$$f_{\text{new}}(\phi; bu, T) = C_0(1 - p(\phi)\eta) + a_{wg}W_b q(\phi)$$

Both C_0 and η come from the same $\text{hbs_state}(bu, T)$ that produces Θ , r_n and X .



Left: Eq. (7) at three stored energies. Right: f_{new} below, at, and above the threshold.

Tandogan *et al.* (2026): nucleation *from* the free energy

- Recrystallization is usually modelled with an imposed nucleation (Muramatsu);
- Tandogan *et al.*⁸ obtain it spontaneously, from stored dislocations at grain boundaries through a one thermodynamically consistent field theory, Cosserat crystal plasticity + Henry–Mellenthin–Plapp (HMP) orientation phase field (HMP-CCP).

Order parameters

- $\eta_T \in [0, 1]$: coarse-grained **crystalline order**, = 1 in the grain bulk, < 1 inside the diffuse GB.
- θ : lattice orientation, identified with the Cosserat microrotation
- 2D, small strain: dofs $(\eta_T, \theta, u_1, u_2)$, FEniCS + Mfront.

⁸Tandogan, Budnitzki, Sandfeld, *J. Mech. Phys. Solids* 206 (2026) 106325. Presented here up to §3.2.

The HMP-CCP free energy and the coupling function $\phi(\eta_T)$

$$\psi = f_0 \left[\alpha V(\eta_T) + \frac{\nu^2}{2} |\nabla \eta_T|^2 + \mu^2 g(\eta_T) |\nabla \theta|^2 \right] + \frac{1}{2} \mathbf{e}^e : \mathbb{E} : \mathbf{e}^e + 2\mu_c (\check{\mathbf{e}}^e)^2 + \underbrace{\phi(\eta_T) \sum_{\alpha} \frac{\lambda}{2} \mu b^2 \rho_{\alpha}}_{\text{stored dislocations}}$$

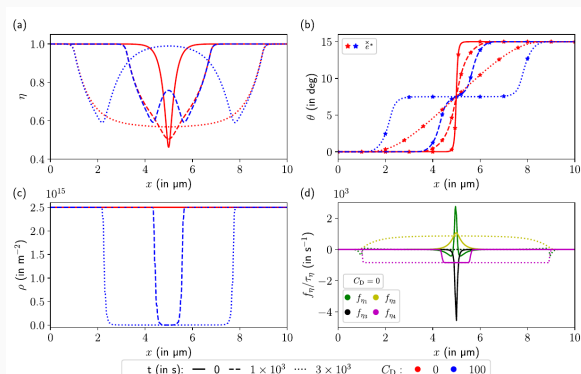
- $V = \frac{1}{2}(1 - \eta_T)^2$: single well, penalises GBs.
- $g(\eta_T)$ singular at 1, with a $\ln$ term $\Rightarrow$ **Read-Shockley** GB energy.
- $\phi, \eta = \frac{c_3}{2} [1 - \tanh(c_1(\eta_T - c_2))]$: a **step**. The stored energy pushes on η_T only inside the GB ($\eta_T < c_2$); the bulk is “crystalline enough”.
- Recovery (Kocks–Mecking): extra sink $-\rho C_D A(\eta_T) \dot{\eta}_T$, active only where $\dot{\eta}_T > 0$: **behind a moving GB**.

The nucleation mechanism: a Cu bicrystal

Periodic bicrystal, $\Delta\theta = 15^\circ$, uniform $\rho_0 = 2.5 \times 10^{15} \text{ m}^{-2}$ switched on at $t = 0$.

$C_D = 0$, **no recovery**: the stored-energy force widens the GB into a non-localised orientation gradient, an *unstable* state.

$C_D > 0$: recovery at the GB centre: a dislocation-free nucleus bounded by two GBs that move outward.



Tandogan *et al.*, Fig. 4: η_T , θ , ρ and the forces on η_T ; red $C_D = 0$, blue $C_D = 100$.

Summary

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Backup — the equations

$$(1) \quad \rho_{\text{tot}}(bu) = 10^{2.2 \times 10^{-2} bu + 13.8}$$

$$(2) \quad G(T, P, x, q), \quad \nu(T, P, x, q) \quad \text{NEA/NSC/R(2024)1 p. 124;} \quad f(\nu) = \frac{1-\nu/2}{1-\nu}$$

$$(3) \quad \rho_{\text{LAGB}}^{\max} = \left(\frac{3n\theta_{\max}}{\beta b} \right)^2, \quad (S/V)_{\max} = \frac{9n\theta_{\max}}{\beta^2 b}, \quad \left. \frac{\Delta R}{R} \right|_{\max} = \frac{k \rho_{\text{LAGB}}^{\max}}{\rho_{\text{tot}}}$$

$$(4) \quad A_1 = \frac{f(\nu)}{4\pi} \ln \frac{\rho_c^{-1/2}}{b}, \quad A_2 = \frac{f(\nu)}{4\pi} \ln \frac{\rho_{\text{tot}}^{-1/2}}{b}$$

$$(5,6) \quad F = \rho_{\text{free}} A_1 G b^2 + \rho_{\text{ord}} A_2 G b^2 = C_0 + C_2 \eta^2 + C_4 \eta^4$$

$$(7,7b) \quad \eta_{\text{eq}} = \min(\sqrt{-C_2/2C_4}, \eta_{\text{bal}}), \quad \eta_{\text{bal}} = \sqrt{\min(\rho_{\text{tot}}/\rho_{\text{LAGB}}^{\max}, 1)}$$

$$(8) \quad \Theta = \min(\eta_{\text{eq}} \theta_{\max} \cdot \frac{180}{\pi}, \theta_{\text{HAGB}})$$

$$(9) \quad r_n = \min\left(\frac{1.5}{(S/V)} \left(1 + \frac{\Delta R}{R}\right), R_{\text{grain}}\right)$$

$$(10) \quad X = \text{clip}\left(\frac{\Theta - \theta_u}{\theta_{\text{HAGB}} - \theta_u}, 0, 1 - 10^{-9}\right)$$

Backup — regimes and parameters

Regimes (T -, P -, x -independent)	
condition	bu [GWd/tU]
$C_2 = 0$, Θ leaves 0	18.88
$\Theta = \theta_u$, X leaves 0	42.99
$\Theta = \theta_{\text{HAGB}}$	108.07

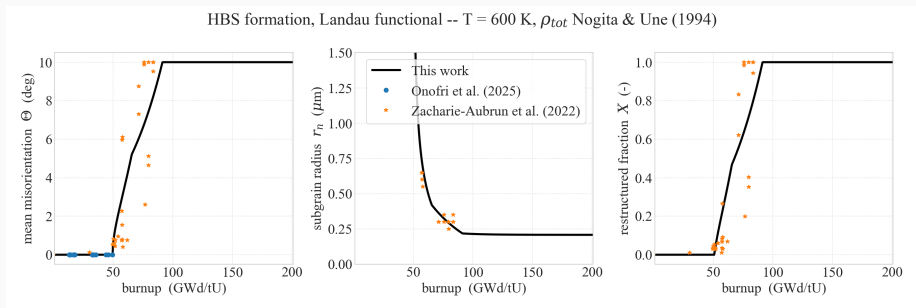
With the set-B parameters the balance bound of Eq. (7b) **never binds**: η is always the stationary point of Eq. (7).

Fixed constants

b	3.889e-10 m
θ_{max}	$= \theta_{\text{HAGB}}$
θ_{HAGB}	10°
θ_u	1.00°
n	2

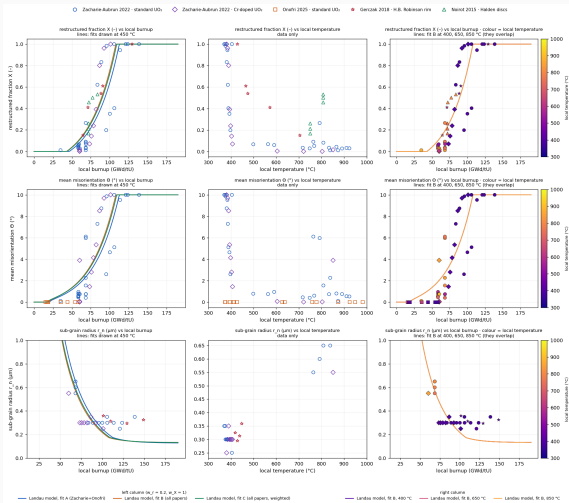
θ_u is the lower end member of the mixture Eq. (10) inverts, and it is set by the **measurement convention**, not fitted: the EBSD data reports a restructured fraction at 1° (f_1) and one at 10° (f_{10}), so 1° is the edge below which a boundary is not counted at all. A map whose whole resolved population sits there carries no restructuring — which is $X = 0$. It fixes only *where* X leaves zero: Θ and r_n do not depend on it.

Backup — the model on its own, $T = 600$ K, $P = 0.05$



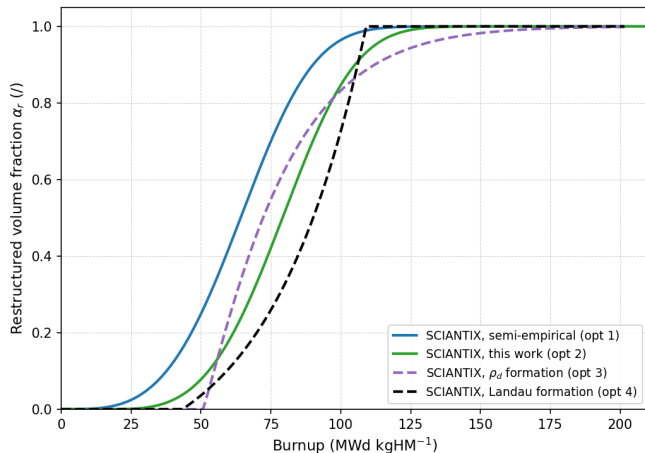
```
python3 hbs_formation_landau.py --plot
```

Backup — fits A, B, C against the data



```
python3 calibrate.py --study --weight 0.2 --fraction-weight 1
```

Backup — the four regression cases, as they ship



Not the controlled comparison of the previous figure: each case pairs its formation option with the porosity model it was built around (1, 2, 3, 3), so part of the spread is the porosity model.

This is what a user of each regression case actually gets;
formation_options.png is what isolates the formation model.

Backup — the four formation options in a full run

HBS formation options on the same irradiation, porosity model 3 throughout

